
ASSESSING PUBLIC TRANSPORT INFRASTRUCTURE: THE ROLE OF EMPLOYMENT MATCHING IN SPATIAL ACCESSIBILITY MEASURES

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Abstract

The definition of accessibility emphasises the role of the attractiveness of opportunities at potential destinations that residents consider valuable. This study revises the common assumption in empirical studies that residents are equally attracted to all types of employment and examines its implications for public transport evaluation. Additionally, the role of the modifiable areal unit problem is also explored in this relationship.

Keywords Accessibility · Employment matching · Public transport infrastructure · Modifiable areal unit problem

1 Introduction

Location-based accessibility measures are valuable for assessing the benefits enabled by transport infrastructure, among others, those in the property market and social equity. At its core, accessibility concerns the ease of travel to opportunities that people consider valuable. This study revises the implications of the common assumption in empirical studies that residents are equally attracted to all types of employment. We focus on employment because it is one of the most frequent purpose for travelling. Additionally, the role of the modifiable areal unit problem (MAUP) is examined in this context. These considerations can provide valuable

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guidance for measuring accessibility in environmental valuation studies or for equity assessments of public transport infrastructure, for example. Thus, the following questions are examined:

1. What are the key spatial differences between generic and matching accessibility measures and what is the role of the MAUP?
2. How does the evaluation of PT infrastructure differs when accounting for matching employment opportunities in accessibility measures?
3. What are the social equity implications of the above?

This paper contributes to literature by:

- Examining the implications of relaxing the assumption that all employment opportunities are equally attractive to all the population in accessibility measures drawing on an empirical case for a ten-year period.
- Examining the role of the MAUP in these effects.
- Reports the findings of GMC, an understudied mega-city, by identifying and digitally reconstructing in a common format (general transit feed specification, GTFS) a series of temporal stages of the MPTN in GMC for a ten-year period (2009-2019).

2 Literature review

Thériault, Voisin, and Des Rosiers (2013) conceptualize accessibility as “the ability that persons living in that area can, within an acceptable travel time, reach a satisfactory diversity of activity locations that are important to them” (p. 234). Levinson and Wu (2020) refer to accessibility measures simply as “the ease of reaching valued destinations” (p. 130).

2.1 Accessibility measures

Location-based measures consist of the sum of the product of two functions, namely $g(\cdot)$ and $f(\cdot)$, as shown in Equation 1 (Páez, Scott, and Morency 2012):

$$A_{ik}^p = \sum_j g(W_{jk})f(c_{ij}^p). \quad (1)$$

Here, accessibility A to opportunities of type k is viewed from location i , from the perspective of person type p . The measure reflects the attractiveness of opportunities W of type k at destination j . The term c_{ij} refers to a function of travel cost between i and j , as experienced by person p .

Equation 1 is often used in literature to measure accessibility to employment as following:

$$A_i = \sum_{j=1}^N E_j f(c_{ij}) \quad (2)$$

The attractiveness of a destination W is expressed by E , usually as the number of employment opportunities at j , i.e. E_j . In $f(c_{ij})$, travel cost is represented with some form of travel time.

In Equation 2, households and opportunities are assumed to be homogeneous, i.e., all types of employment are equally attractive for all types of residents. However, contributions from theory highlight the characteristics of the demand (residents at origins) and supply at potential destinations (Geurs and van Wee 2004; Levinson and Wu 2020). Specifically, Levinson and Wu state that “different job categories represent different economic activities, and different types of jobs generally require different skills and education levels” (2020, 147). In the context of the labour market, this is supported by the Human Capital Theory (Schultz 1961; Becker 1994).

Location-based measures can consider these elements. For instance, Equation 1 refers to the following components: (1) individual’s profiles; 2) how different profiles perceive or experience the space, and; 3) constraints that may prevent segments of the population to access some type of activities. Some of these aspects have been found to be relevant in empirical studies. For instance, gender in Osland (2010) or age and gender in Thériault, Voisin, and Des Rosiers (2013) in the study of the property prices.

The operationalization of these measures can vary. Levinson and Wu (2020) suggest to weight employment in each category with the number of eligible workers. In empirical studies, these characteristics can be expressed as type of person m at j and type of employment m at k (Pereira, Grégoire, and Karner 2019). Accordingly, the measure can be specified as following:

$$A_{im} = \sum_{j=1}^N E_{im} f(c_{ij}), \quad (3)$$

Here, only employment opportunities E matching the profile of residents between j and k are considered in the accessibility measure. Thus, this reflects the notion that residents would find attractive only employment opportunities that match their skills and education level with corresponding employment opportunities.

2.1.1 Evaluation of public transport

Location-based accessibility measures have been used to evaluate the performance of public transport systems in policy report and research (Ermagun 2021; Liu, Porr, and Miller 2022; Conway, Byrd, and Van Eggermond 2018; Malekzadeh and Chung 2020; Palmateer et al. 2021). A measure based on realistic travel times can reflect important elements of a public transport system. This is because travel time by public transport is affected by several aspects of the system, such as:

- The topology of the network, e.g. connectivity, centrality.

- Capacity of the infrastructure, e.g. degree of right of way segregation, capacity and design of stations.
- Spatial coverage, e.g. availability of infrastructure and catchment areas.
- Operational characteristics, e.g. frequency of services, overlapping routes, express services, transfer wait times.

Despite the growing relevance of accessibility measures in the evaluation of public transport infrastructure, knowledge of the implications of overlooking matching employment are limited in literature.

3 Methods

3.1 Accessibility measures examined

The general accessibility is computed as following for this work:

$$A_{jt}^{PT} = \sum_{j=1}^J E_k f(d_{jkt}) \quad (4)$$

The implementation of matching accessibility is adapted according to the information available for the area of study and some preliminary findings. The households' education level (i.e., median number of years of education) is employed assuming correspondence with the level of salary paid in the employment market. This is further supported by the positive relationship found between education level and wages in empirical studies in the context of Mexico (Nuñez, Paredes, and Garduño-Rivera 2017; Rojas, Angulo, and Velázquez 2000).

$$A_{jtm}^{PT} = \sum_{j=1}^J E_{km} f(d_{jkt}), \quad (5)$$

Here, m at j is categorized according to the median education level of its residents by quintile and m at k is classified based on the average paid salary by quintile. Only employment opportunities E matching the education level quantile between j and k are considered (when $j \neq k$).² This assumes that residents would deem valuable only employment opportunities that match their education level, i.e. those in the same quintile. The data sources and specific procedures to process the data are described in the next sub-section.

3.2 Area of study

Greater Mexico City (GMC) (*Zona Metropolitana del Valle de México*) is located in in the south-centre of Mexico. In 2015, GMC had nearly 21 million inhabitants (INEGI 2015b). The official delimitation consists of 76 adjacent municipalities, which are part of three different states, namely: (1) *Ciudad de México* (16

²An exception occurs when $j = k$. In this case, employment opportunities in the origin are considered as available. This is to avoid the assumption implying that some zones have accessibility values equal to 0

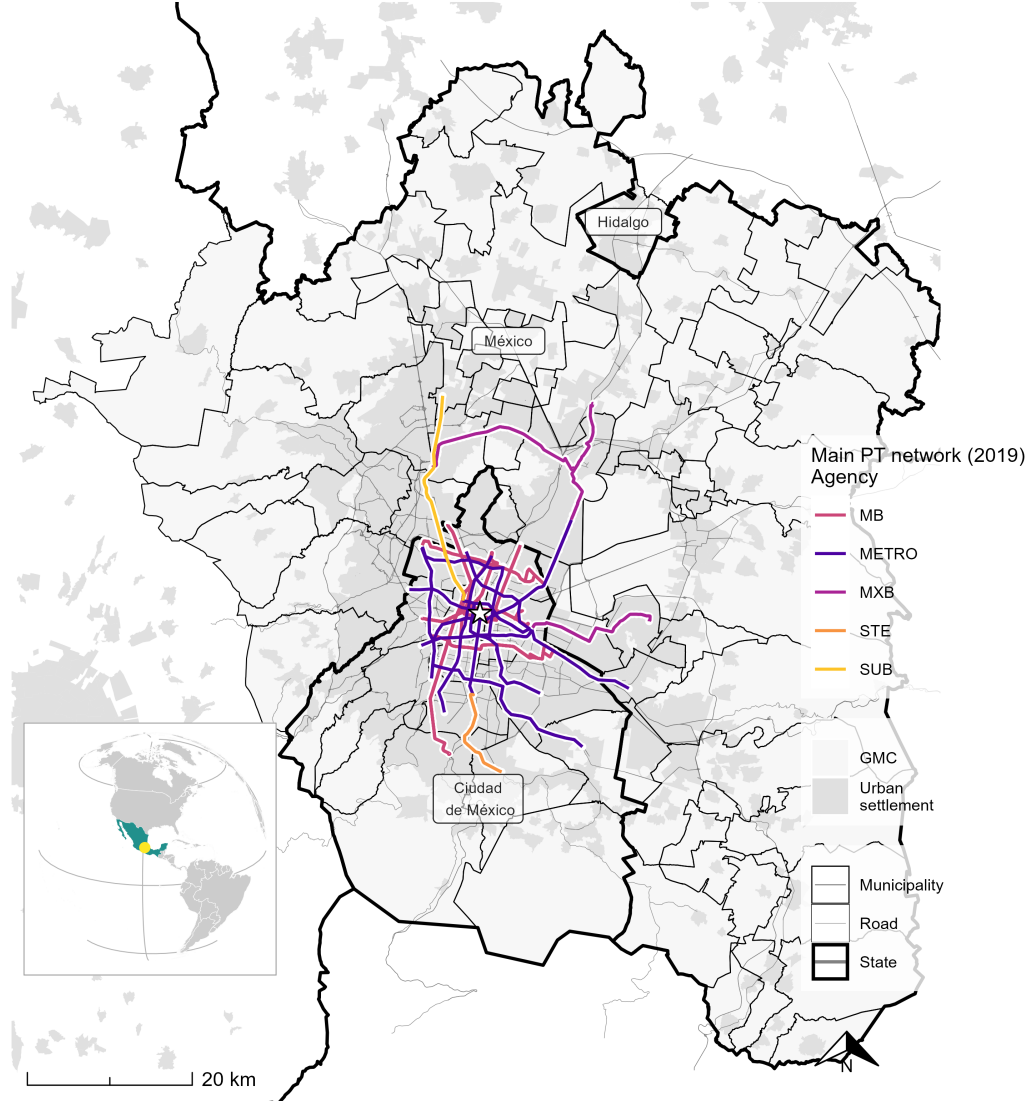


Figure 1: Main public transport network (MPTN). Greater Mexico City, 2019.

Source: Verduzco-Torres (2023) based on Marco Geoestadístico Nacional 2014 Versión 6.2 (INEGI, 2014); Delimitación de las zonas metropolitanas México 2015 (INEGI, 2018); Secretaría de Movilidad de la Ciudad de México (SEMOVI, n.d.); Sistema de Transporte Masivo y Teleférico del Estado de México (SITRAMyTEM, n.d.); OpenStreetMap (OSM, n.d.).

municipalities); (2) *Hidalgo* (1 municipality), and (3) *México* (59 municipalities), as shown in Figure 1. The total GMC area is 7,889 km², and it is home to 17% of the national population.

3.2.1 The main public transport network (MPTN)

The focus of this work is on *rapid transit* and *semi-rapid transit* infrastructure, excluding specialized modes i.e. *aerial tramway*. The combination of public transport modes included in the analyses is referred to as the *main public transport network* (MPTN). Figure 1 shows the MPTN in GMC for the year 2019. This consists of a variety of modes operated by the following main agencies: The metro system, directly operated by the

Table 1: Main public transport network extensions between 2010 and 2019 in Greater Mexico City grouped by temporal stage.

Corridor	Agency	From station	To station	Opening date	Previous change*	GTFS source
Stage 7: 2018 to 2019						
7	MB	Indio Verdes	Campo Marte	2018-03-01	26	2019-01-09
Stage 6: 2016 to 2018						
6	MB	El Rosario	Martín Carrera	2016-01-01	12	2017-09-02
Stage 5: 2015 to 2016						
2	MXB	La Quebrada	Las Americas	2015-01-01	14	2015-01-28
Stage 4: 2013 to 2015						
5	MB	Río de los Remedios	San Lázaro	2013-11-01	6	2014-12-30
3	MXB	Chimalhuacán	Pantitlán	2013-05-01	7	2014-12-30
Stage 3: 2012 to 2013						
12	Metro	Mixcoac	Tlahuac	2012-10-01	6	2013-10-21
4	MB	Buenavista	San Lazaro Aeropuerto	2012-04-01	14	2013-10-21
Stage 2: 2010 to 2012						
3	MB	Tenacayuca	Etiopía	2011-02-01	4	2013-10-21
1	MXB	Ciudad Azteca	Ojo de Agua	2010-10-01	10	2013-10-21
Stage 1: Before 2010						
2	MB	Tepalcates	Tacubaya	2009-12-01	NA	2013-10-21

Note:

*Months relative to the previous modification of the main public transport network.

state-owned company *Sistema de Transporte Colectivo* (STC) and designated as Metro; two BRT systems, namely *Metrobus* (MB) in Ciudad de México and Mexibus (MXB) in the state of México; One LRT line operated by STE; and one regional rail line designated as *Sub*.

The period of study was divided into seven temporal stages to reflect significant changes in the MPTN over time. Initially, changes were identified by the physical alterations in the MPTN, such as the opening of a new corridor or the extension of an existing one. Table 1 details these changes and their corresponding temporal stages from 2010 to 2019. The stages were grouped based on the time interval between changes, with modifications occurring within six months being grouped together in the same stage. This criteria follows previous empirical studies and assumes that accessibility benefits are perceived as simultaneous when changes happen within short timeframes (Ahlfeldt 2013; Gibbons and Machin 2005).

The MPTN stages and corresponding corresponding corridors are in Figure 2.

3.2.2 Spatial analytical framework

Various spatial analytical schemes (SAS) are considered to address the research questions related to the MAUP. The area of study is systematically partitioned in uniform cells, j , of various gradual sizes to examine the scale effects, drawing on Wong, Lasus, and Falk (1999). The criteria and implementation employed is

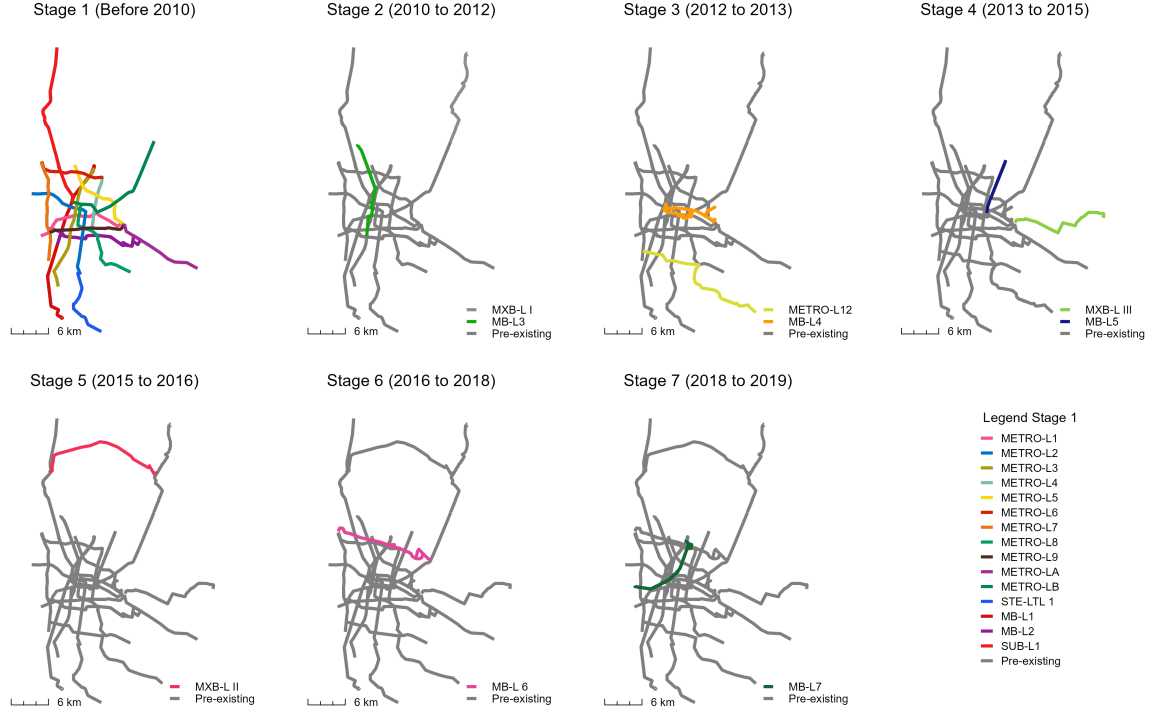


Figure 2: Temporal stages of the main public transport network. Greater Mexico City, 2010 to 2019.
Source: The author based on Semovi (n.d.) and SITRAMyTEM (n.d.).

Table 2: Spatial analytical schemes (SAS) overview.

	Grid 0.5 km	Grid 1 km	Grid 2 km	Grid 4 km	Post code
Count	13 297	3 972	1 264	413	2 377
Mean area (Sq. km)	0.22	0.86	3.44	13.55	1.94
Mean population (Inhabitants)	1 498.13	5 093.21	16 324.62	50 187.76	8 317.75

Source:

Censo de Población y Vivienda 2010 (INEGI, n.d.)

Sistema para la Consulta de Información Censal 2010 (INEGI, n.d.)

Mapa de Ubicación de Códigos Postales (V5). Correos de México, 2019

adopted from empirical accessibility studies, using hexagonal cells of 0.5 km, 1 km, 2 km, and 4 km (Pereira 2018, 2019). This is to (1) maintain consistency in this research area, (2) allow comparability, (3) exploit existing analytical tools and resources and, (4) augment previous findings in a different context. A further scheme defined by the official post code zones was considered to account for zoning.

Table 2 summarises the main characteristics of each SAS for cells containing residents or economic activity. The total number of cells ranges from 13,300 in the highest resolution grid (0.5 km) to 410 in the lowest (4 km), with surface areas varying between 0.22 km² and 13.5 km². On average, the smallest cells contain only 1,500 inhabitants, while the largest have over 50,000.

The polygons were created manually in R programming language. Post code polygons are sourced from national Post Office (*Correos de México*) ‘Version 5’.

3.3 Data

3.3.1 Demographic

The demographic comes from the 2010 Population Census (INEGI 2012). These data were selected for their detailed disaggregation at the city block level, including key fields such as total population and average years of education completed by individuals aged 15 and older.

The count data included were transferred to the SAS using the *areal weighting method* (Saporito et al. 2007). This reassigns the data according to the proportional overlay of the source and target areas. The average education level was assigned using the *population weighting method*. In addition, this takes into consideration the number of inhabitants and their type.

3.3.2 Employment

Employment data are sourced from the 2014 Economic Census (*Censos Económicos 2014*) (INEGI 2015b) and accessed at the city block level via the Microdata Lab and Remote Processing service (<https://www.inegi.org.mx/microdatos/>). The dataset includes: 1. Number of establishments (fixed locations producing goods or services); 2. Employed personnel (individuals active in establishments, including volunteers and family members), referred to as ‘employment’; and 3. Total gross remuneration paid for the relevant year.

Employment information which came aggregated at higher geographic level than the city-block due to privacy (<8%) were re-allocated using proportional estimates based on the spatial distribution from the 2015 DENU (INEGI 2015a). This is an open-access, point-level dataset of individual establishments. The employment data was assigned from city blocks to SASs following the same approach employed for the demographic data, i.e. according to the areal weighting method.

The average income was computed for each areal unit based on the total number of employment and the total gross remuneration paid.

3.3.3 Public transport timetables

The estimated travel time used in accessibility measures is based on scheduled public transport services. The Transport Authority in Mexico City (*Secretaría de Movilidad de la Ciudad de México*, Semovi) released the first *general transit feed specification* (GTFS) files in 2013, which were accessed via the Open Mobility website (<https://openmobilitydata.org/>). These datasets cover the agencies operating most MPTN services in GMC. However, there are some limitations: firstly, the geographic coverage primarily includes services within

Ciudad de México, excluding BRT services provided by MXB. Secondly, the temporal coverage starts in 2013, with updates occurring only sporadically, approximately once a year.

To overcome the first limitation, the geographic coverage of the official GTFS data was augmented to include three BRT corridors operated by MXB. This was carried out *manually* based on unstructured information provided by the Transport Agency of the State of Mexico (*Sistema de Transporte Masivo y Teleferico del Estado de México*, Sitramytem). This corresponded to services for 2019 only.

Regarding the second limitation, the MPTN’s temporal stages were reconstructed using the GTFS standard. The primary input for each stage was the closest available official GTFS data reflecting changes in the MPTN, with only the relevant services for each corridor in their respective stages being retained. For MXB services, 2019 data was considered due to the lack of earlier details, assuming uniform characteristics of the BRT corridors across the stages. Similarly, for other corridors included in early stages, services were assumed to reflect 2013 levels due to the unavailability of earlier data. All non-MPTN services were excluded, thereby focusing exclusively on the benefits of the MPTN.

3.3.4 Road network and pedestrian infrastructure

The road and pedestrian infrastructure network is sourced from *OpenStreetMap* (OSM) (Manually downloaded on 30th October 2019 through a web export tool available in <https://export.hotosm.org/es/v3/>). This enables multi-modal journeys to realistically represent the access/egress stage to a public transport service when combining walking and public transport.

3.4 Travel costs and impedance parameters

3.4.1 Modeled travel time

Travel time matrices were calculated for each origin-destination (OD) pair across SASs using a combination of walking and public transport within the main public transport network. OD points i and j are represented by population-weighted centroids of spatial units containing economic activity (from the 2014 Economic Census) or demographic data (from the 2010 Population Census). Travel times are capped at 180 minutes, based on patterns observed in the 2017 Travel Survey.

Travel times were modelled using a door-to-door approach with OpenTripPlanner (OTP) version ‘1.4.0’ (OTP n.d.) via the implementation of an open-access routine written *Python* (Pereira et al. 2019). This approach considers access, waiting, onboard, transfer, and egress times. All-to-all travel time matrices were generated for each SAS across seven temporal stages using reconstructed GTFS files and road/pedestrian networks, with eight random departure times between 10 a.m. and 2 p.m. The median travel time was used for accessibility models (Conway, Byrd, and van der Linden 2017). This process, computationally intensive due to the high resolution of SASs, resulted in 2.1 billion routes. Matrices were then merged, and median

travel times calculated, with walking distances limited to 3 km when using public transport and 4.8 km (60 minutes) for walking-only trips.

The internal-zone travel-times were estimated based on the surface area as following: $d_{i=j} = \frac{1}{2} \sqrt{\frac{\text{area}_i}{\pi}}$ (Frost and Spence 1995; Stepniak and Rosik 2015). The metric distance was transformed to travel time using the average speed modelled for short journeys (less than 3 km).

3.4.2 Spatial decaying function and calibration of accessibility parameters

The spatial decaying function and parameters for accessibility measures were determined using estimates from a trip distribution model, following Handy and Niemeier (1997). This model is used to understand how commuters in Greater Mexico City (GMC) discount employment opportunities based on travel time. The analyses use data from the 2017 Travel Survey, focusing on a weekday and public transport combined with walking.

Alternative specifications were estimated within a doubly constrained gravity model framework, expressed as following:

$$T_{jk} = A_j O_j B_k D_k f(d_{jk}) \quad (6)$$

where

$$A_j = \left[\sum B_i D_i f(d_{jk}) \right]^{-1} \quad (7)$$

and

$$B_k = \left[\sum A_k O_k f(d_{jk}) \right]^{-1} \quad (8)$$

with balancing factors A_j and B_k calculated iteratively. The model estimates commuter flows (T_{jk}) between origins (j) and destinations (k), with constraints on the number of journeys at each. $f(d_{jk})$ is a spatial deterrence or decaying function. The inverse power ($d_{jk}^{-\beta}$) and negative exponential ($e^{-\beta d_{jk}}$) decaying functions were tested. β represents the deterrence parameter.

A modified specification intended to capture an add-up cost for inter-zone journeys or analogously a benefit of residing and working within the same zone is considered (Thorsen and Gitlesen 1998). This includes a parameter μ which affects only the diagonal elements of the OD matrix. The term is given by the following expression:

$$\mu \delta_{jk} \quad (9)$$

where Kronecker delta takes the following value:

$$\delta_{ij} \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k \end{cases} \quad (10)$$

The simultaneous calibration of the model parameters β and μ is achieved via optimization by minimizing the standardized root mean square error (SRMSE) using the *Nelder-Mead's simplex algorithm* as implemented in base R (R Core Team 2021).

The preliminary analyses shows that the negative exponential function provided more accurate estimates compared to the inverse power. Also, the use of the within-zone parameter μ improves the estimates. Therefore, the selected specification is the modified negative exponential including μ . The parameters employed to estimate accessibility are shown in Table .

3.5 Analytical framework

3.5.1 Accessibility indicators

$$\text{Similarity rate} = \frac{A_{jtm}^{\text{PT}}}{A_{jt}^{\text{PT}}}$$

3.5.2 Accessibility equity measures

3.5.3 Indicators of spatial association

- Global Moran's I .
- Local indicators of spatial association (LISA). Local Global Moran's I .

4 Findings

4.1 Spatial analysis

Table 2 summarises descriptive statistics for both accessibility measures across various SAS. The mean values for the measure considering all type of employment range from 59,400 in the 0.5 km grid to 107,000 in the postcode schemes, while the matching measure ranges from 25,500 to 41,600. This indicates a smaller scale for the latter due to restrictions imposed in employment opportunities. However, the ratio between the means varies with scale. For instance, the 'All' measure is about 2.3 times larger in the 0.5 km grid and only 1.1 in the 4 km grid, with the postcode scheme at 2.6. Notably, the mean of the matching measure increases steadily with scale, from 25,500 in the 0.5 km grid to 54,900 in the 4 km grid, whereas the 'All' measure shows no consistent trend. The postcodes scheme results in the largest mean values. This can be explained by zoning effects. Specifically, peripheral regions, which are associated with low accessibility, tend to be larger, thereby reducing the number of low-accessibility value units. Meanwhile, equal-sized grids fragment these regions of the city into a larger number of units, thereby pulling the mean lower.

Table 3: Summary statistics of accessibility measures over various SAS.

SAS	All	Matching	N
Grid 0.5 km	59.4 (121.3) [0.9, 50.8]	25.5 (71.4) [0.3, 9.2]	12 535
Grid 1 km	55.1 (120.8) [0.8, 31.7]	28.5 (83.1) [0.3, 10.8]	3 765
Grid 2 km	54.1 (130.4) [0.8, 25.0]	37.1 (108.6) [0.5, 16.6]	1 182
Grid 4 km	62.8 (166.6) [0.9, 29.2]	54.9 (158.1) [0.8, 27.5]	387
Post code	107.1 (149.2) [4.8, 157.3]	41.5 (85.7) [2.1, 33.5]	2 301

Note:

Descriptive statistics: Mean (SD) [1st-Q, 3rd-Q]. Figures shown in thousands.

The series of maps in Figure 3 show the modelled MPTN accessibility to employment aggregated at various SASs (columns) by type of accessibility specification (rows). The accessibility values are presented in decile breaks. The top row illustrates accessibility to employment considering all opportunities. High resolution grids, 0.5 km or 1 km, clearly highlight the relevance of both central areas and radial public transport axes or corridors. This reflects an overall star-shaped structure. The post code SAS also reflects some of the detailed patterns partially. Examples of corridors connecting non-central areas include the regional rail Line 1 *Buenavista-Cuautitlán* (SUB-L1) to the northwest and the Metro line B (METRO-LB) combined with the Mexibús BRT Line I (MXB-LI) to the northeast. The SUB-L1 generates high-accessibility “islands” around stations, even far from the CBD. In contrast, the northeast corridor’s effects are most pronounced along the METRO-LB segment, diminishing further from the CBD and in areas served by the BRT MXB-L1. These varying patterns arise mainly from the physical and operational characteristics of the transport modes and land use, i.e. operational speed, distance between stations, and proximity to main employment centres.

Median MPTN accessibility to employment by public transport. Greater Mexico City.
Accessibility is computed as the median of the period of study for each spatial unit.

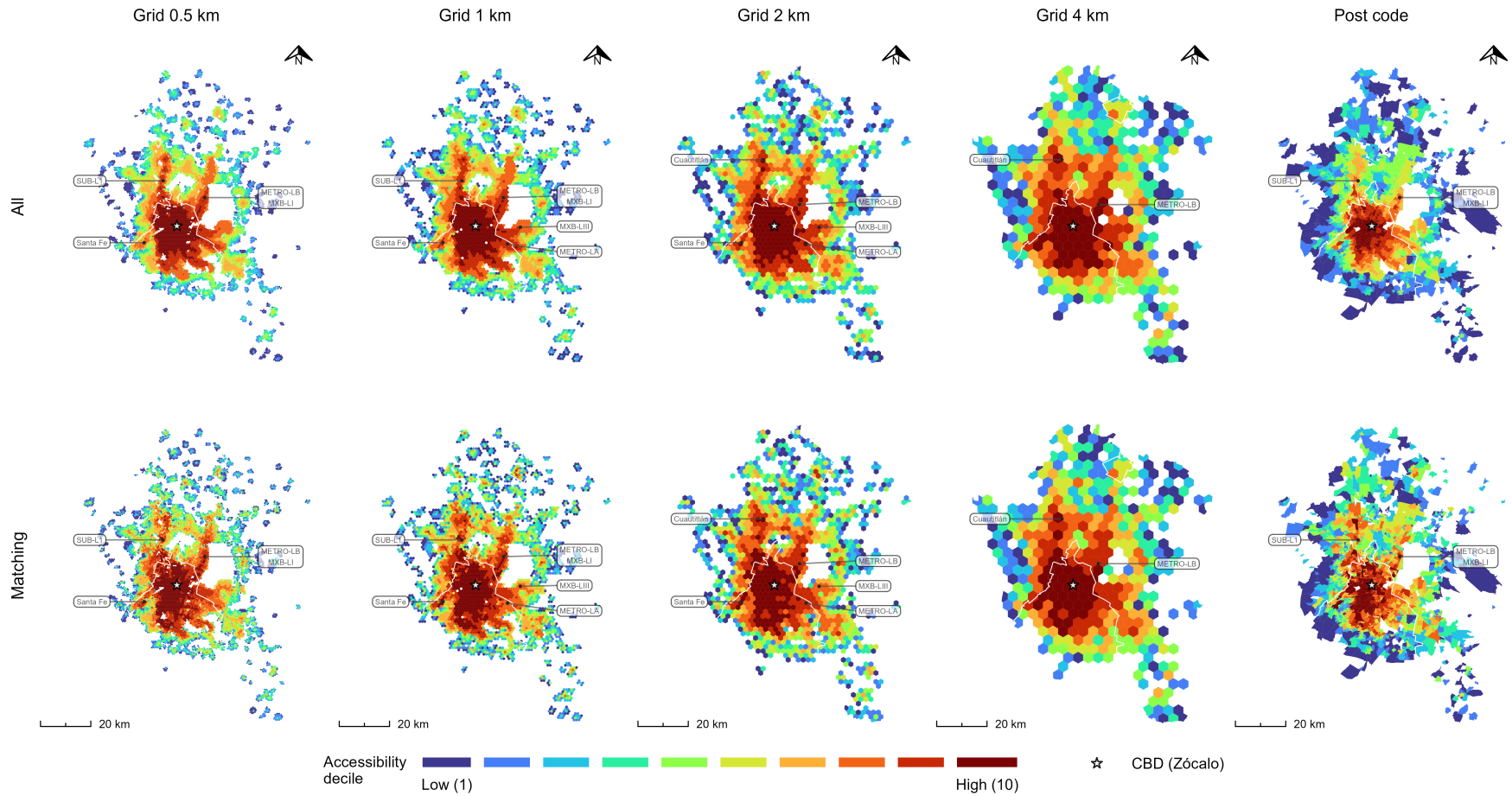


Figure 3: Median MPTN general and matching accessibility over various spatial analytical schemes. Greater Mexico City.
Source: The author based on Semovi (n.d.) and SITRAMyTEM (n.d.).

Table 4: Correlation coefficient over various SAS

Type	SAS				
	Grid 0.5 km	Grid 1 km	Grid 2 km	Grid 4 km	Post code
<i>Bivariate correlation</i>					
Pearson	0.860***	0.879***	0.937***	0.989***	0.780***
<i>Global spatial autocorrelation</i>					
All	0.926***	0.888***	0.812***	0.690***	0.929***
Matching	0.845***	0.795***	0.742***	0.657***	0.737***
Signif. Codes: ***: >0.01, **: >0.05, *: >0.1					
SAS = Spatial analytical scheme.					

Whilst high-resolution SASs reflect detailed characteristics of the MPTN, many of these features are smoothed into polygons or zones in low-resolution schemes, e.g. 2 km or 4 km grid. At this scales, the star-shaped structure develops into concentric-like bands or accessibility rings. At the same time, the differences between the measure are less visible as the scale increases.

The results of a bivariate Pearson correlation test between the generic measure and the matching are presented in Table 4. It is shown that the correlation coefficient increases from 0.86 in the 0.5 km scheme to 0.99 in 4 km one. This shows that the added information by the refined matching measure is lost as the scale of spatial units increases. This is likely because accessibility measures emphasize employment that are in proximity. Therefore, large units mostly capture the within-zone opportunities. The post code schemes show the lowest similarity.

Another general observation from Figure 3 is that the matching measure produces more spatially heterogeneous patterns than the measure considering all types of employment. This is confirmed by the global spatial autocorrelation test results in Table 4. Moran’s I statistic ranges from 0 (no spatial autocorrelation) to 1 (high spatial autocorrelation). The results show that the matching measure consistently produces lower values than the generic accessibility measure. However, the gap narrows as the scale increases. For example, the difference is 0.08 points in the smallest grid but only 0.03 points in the 4 km grid. Postcode boundaries, however, show the largest gap, nearly 0.2 points.

4.1.1 General versus matching accessibility and equity implications

In the next step of the analysis, the accessibility measures are directly compared using a similarity rate, where values close to 0 indicate low similarity and values close to 1 indicate high similarity. We are also interested in whether the differences between matching and generic accessibility measures follow a spatial pattern. Figure 4 displays the local Moran’s I clustering results for the similarity rate across each SAS. High- and mid-resolution SAS reveal a consistent West-East divide, with High-High clusters located towards the West and Low-Low clusters towards the East, using the CBD as a reference point. This suggests that differences between the accessibility measures tend to be stronger in the eastern areas of the city.

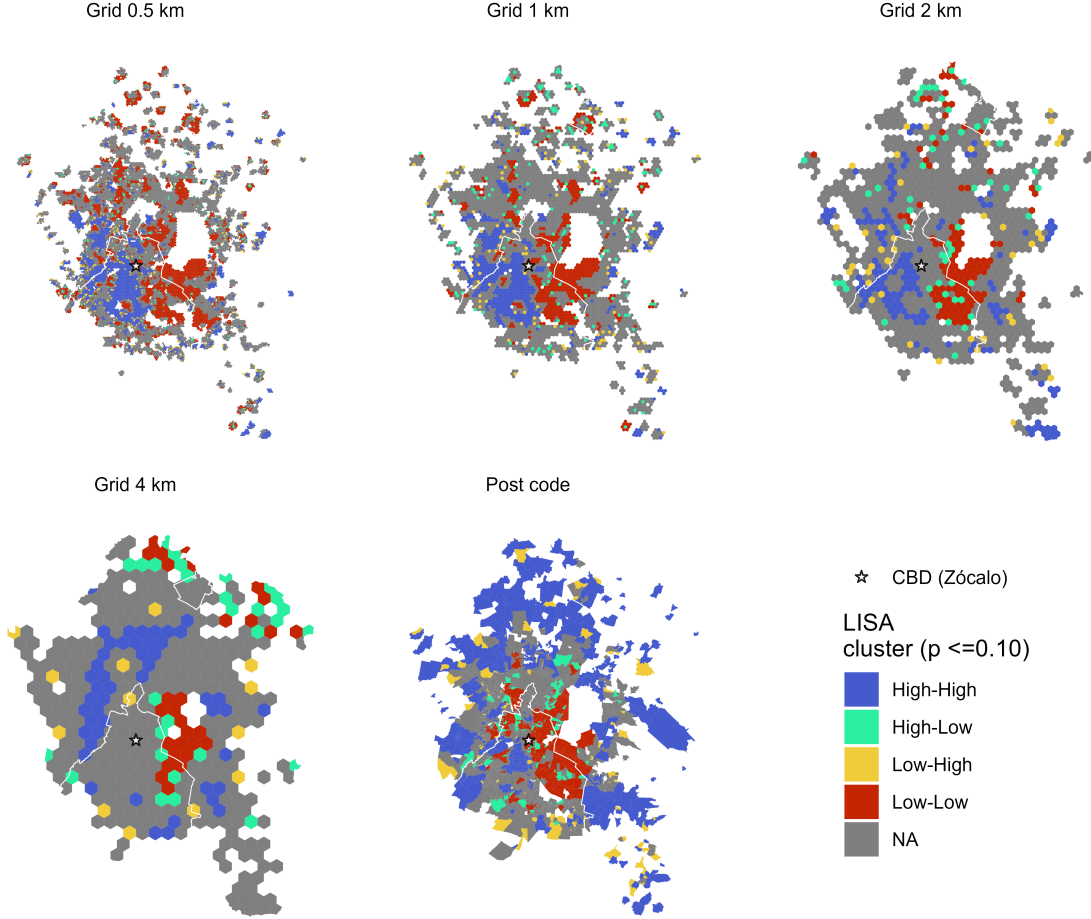


Figure 4: Local indicator of spatial association on similarity rate between generic and matching MPTN accessibility over various SAS.

SAS = Spatial analytical scheme; MPTN = Main public transport network.

While the postcode and 4 km grid show a similar trend for Low-Low clusters, the High-High patterns are less consistent across SAS. In particular, the postcode SAS show substantial differences in the spatial distribution of High-High clusters, with many located in peripheral areas. Additionally, postcode zones in the periphery tend to be large, regarding nearby employment opportunities more important, as travel times to jobs outside the zone are generally long. Since the matching measure accounts for all within-zone employment opportunities, the similarity rate is higher in many peripheral zones. Thus, this is one way in which zoning can interact in the differences between measures.

The overall West-East divide revealed in the LISA analysis is generally consistent with well-known spatial trends of key socioeconomic indicators in GMC, i.e. with socio-economically advantaged population residing in the West (**Ortega-Avila2023?**). To examine potential systematic differences, the LISA clusters are examined by their social composition. Figure 5 displays the proportion of the population which were classed in High-High or Low-Low clusters by education quintile for all SAS. This only includes significant clusters. It

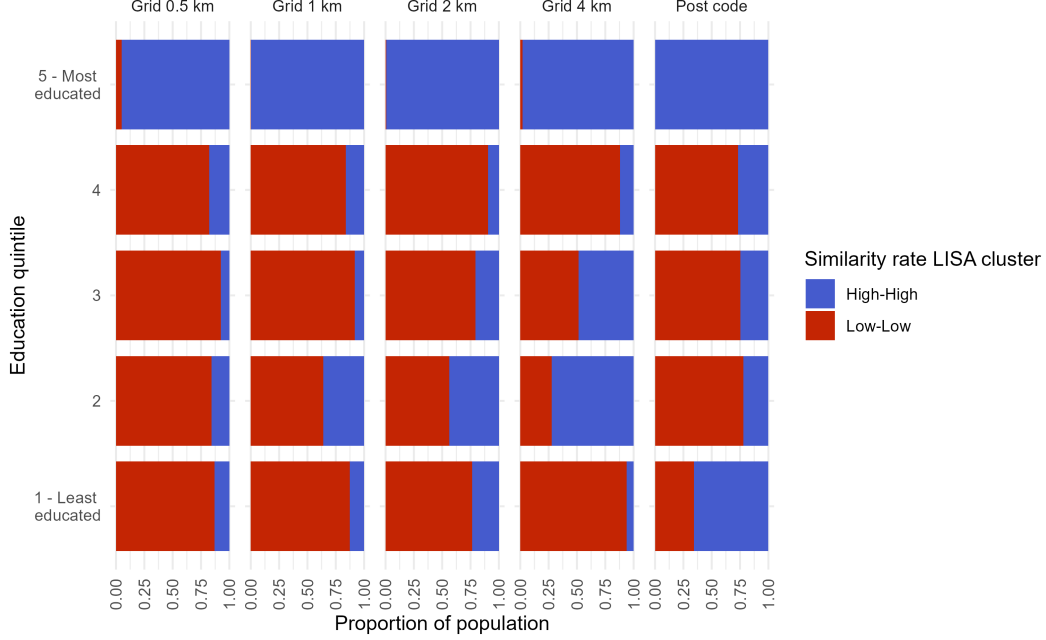


Figure 5: Composition of local indicator of spatial association clusters on similarity rate by education quintile. LISA outlier clusters are not shown.

is clear that the accessibility measures are very similar for the vast majority of the most educated population across all SAS. By contrast, most of the population in less educated groups fall within Low-Low clusters.

The post code scheme shows an exception to this trend. Here, the proportion of the population in with high similarity in is higher. Likely, this is related to the characteristics presented in the previous paragraph, where periphery zones tend display high similarity rates because of the areas of these zones is large. At the same time, these areas populated by low-education groups to a good extent, particularly to the North and North-East.

4.2 Main public transport network assessment

In this section, the focus shifts from spatial variations to temporal changes, aiming to capture the effects of public transport network extensions. To better distinguish these changes, the spatial unit of aggregation is kept fixed. Therefore, the following analysis uses only the 0.5 km grid, as it is the smallest and most detailed SAS.

Figure 6 provides an overview of the relative average changes in the network weighted by the population across various temporal stages, using the distinct accessibility measures. The baseline is set to the ‘Before 2010’ period. Generally, both measures show consistent trends, with notable increases in accessibility during the Stage 3 and Stage 7. These include the introduction of a 25-kilometre metro line (METRO-12) connecting south-east zones to the MPTN, and a BRT line (MB-L7) in a central area of the MPTN, respectively. Extensions in other stages show minor average increases.

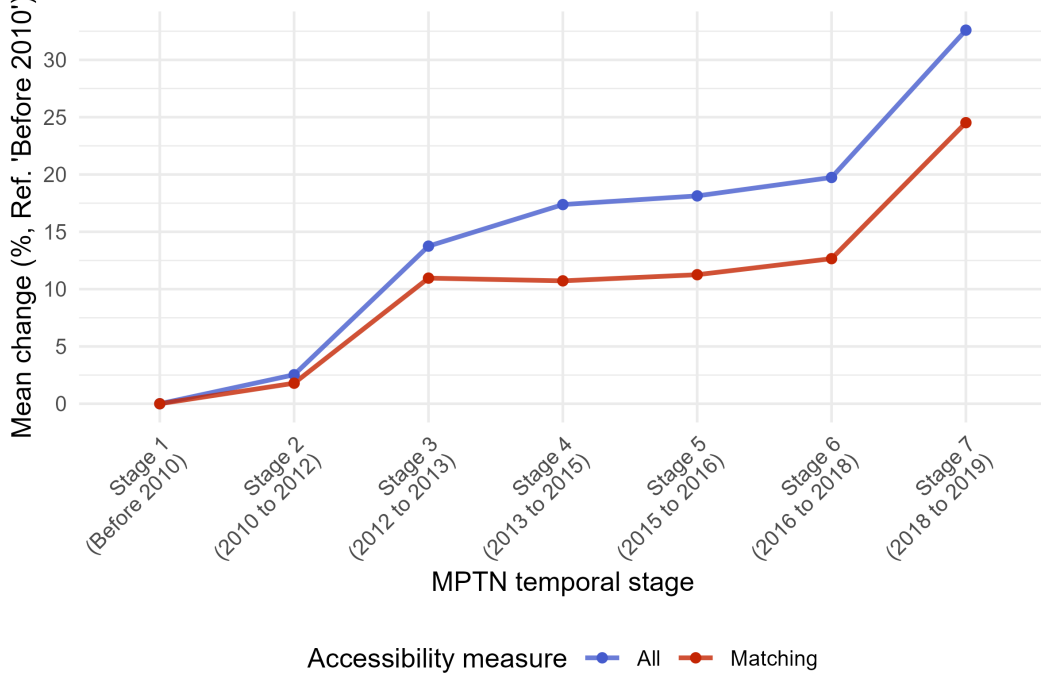


Figure 6: Relative change of accessibility population weighted mean over various temporal stages.
 <!-- ... -->

Interestingly, the general accessibility measure in Figure 6 shows moderate increases in Stage 4 (2013-2015) and Stage 5 (2015-2016), while the matching measure records little to no change, and even a slight decline during these stages. This divergence suggests that while the infrastructure improvements introduced in these periods allowed access to additional employment from certain residential zones, the jobs at destinations did not match the educational qualifications in the areas of influence.

We also draw on the results from a transport equity perspective. An advantage is that it can distinguish between populations that were already served at an ‘adequate’ level before an intervention and those who were not. Figure 7 shows the head count ratio which are under various minimal accessibility thresholds over the MPTN temporal stages for the ‘All’ and ‘Matching’ measures. From the plot, a general reduction in the population under the accessibility thresholds can be observed. This is in the direction expected.

There are additional insights to draw from the results. First, the scale of the ratio is smaller for the matching measure, with the gap narrowing as the percentile increases. This relates to the different density distributions of the measures, where the ‘Matching’ measure is more skewed than the ‘All’ measure. Still, both measures show parallel trends for most stages, with a few exceptions. For example, the ‘All’ measure consistently declines between Stage 1 (‘Before 2010’) and Stage 4 (‘2013 to 2015’) across all thresholds, while the ‘Matching’ measure only begins to reflect this trend after the 15th percentile, showing similar trends by the 30th percentile.

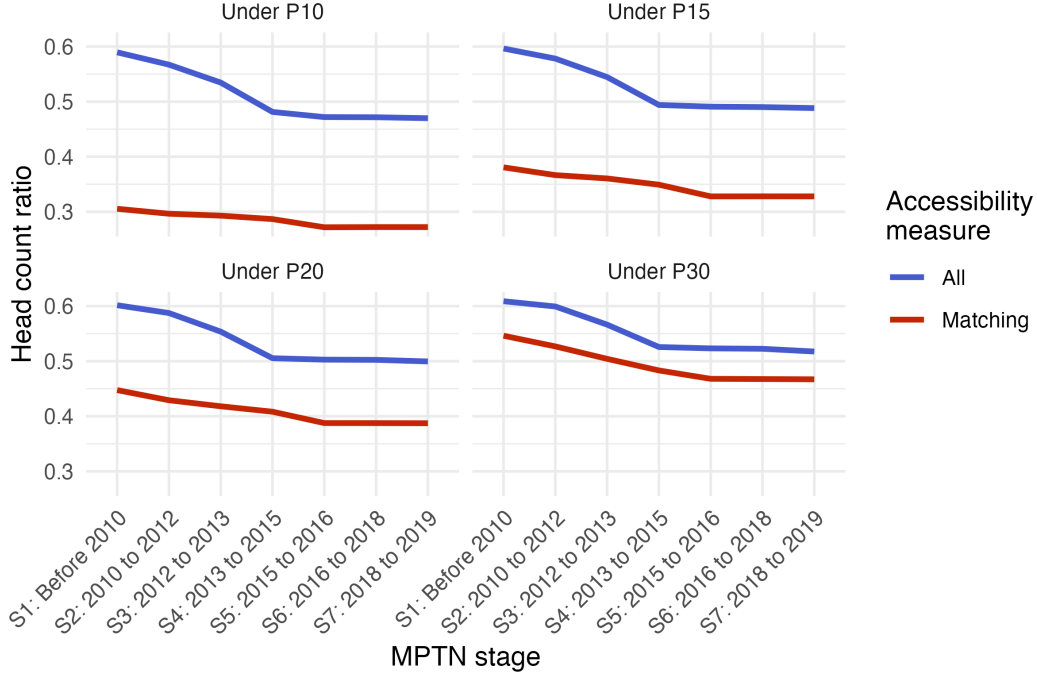


Figure 7: Equity analysis over various temporal stages. S = Stage.
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Another distinction is that the ‘Matching’ measure shows significant improvement in Stage 5 (‘2015 to 2016’) for all thresholds, which is not captured by the ‘All’ measure. This improvement corresponds to the BRT corridor MXB-LII, located in the periphery of the MPTN and connecting key corridors in the northern areas of the city. This suggests that some northern populations were previously considered well-served by the generic measure, even though the employment opportunities did not match their qualifications. This means that the opening of MXB-LII enabled access to matching employment opportunities for those who already had access to a minimum number of general opportunities.

The global population-weighted average in Figure 6 highlights large accessibility benefits introduced in Stage 7 (2018-2019). However, the equity analysis results in Figure 7 show consistent negligible impacts across all percentiles and measures in this stage. This is because the MB-L7, implemented during this stage, is centrally located within the MPTN and primarily serves to articulate existing network corridors. As a result, the additional accessibility to employment was concentrated in areas that already exceeded the examined thresholds.

Figure 8 shows the net population taken out from various accessibility thresholds by education quintile across the MPTN stages. Between Stage 2 and Stage 4, the groups benefiting from the infrastructure are concentrated in the intermediate quantiles, a trend more pronounced in the general accessibility measure. The matching measure remains consistent for some stages, particularly for the 30th percentile threshold. Interestingly, the matching measure reveals a progressive distributional effect in Stage 5, where the benefits go

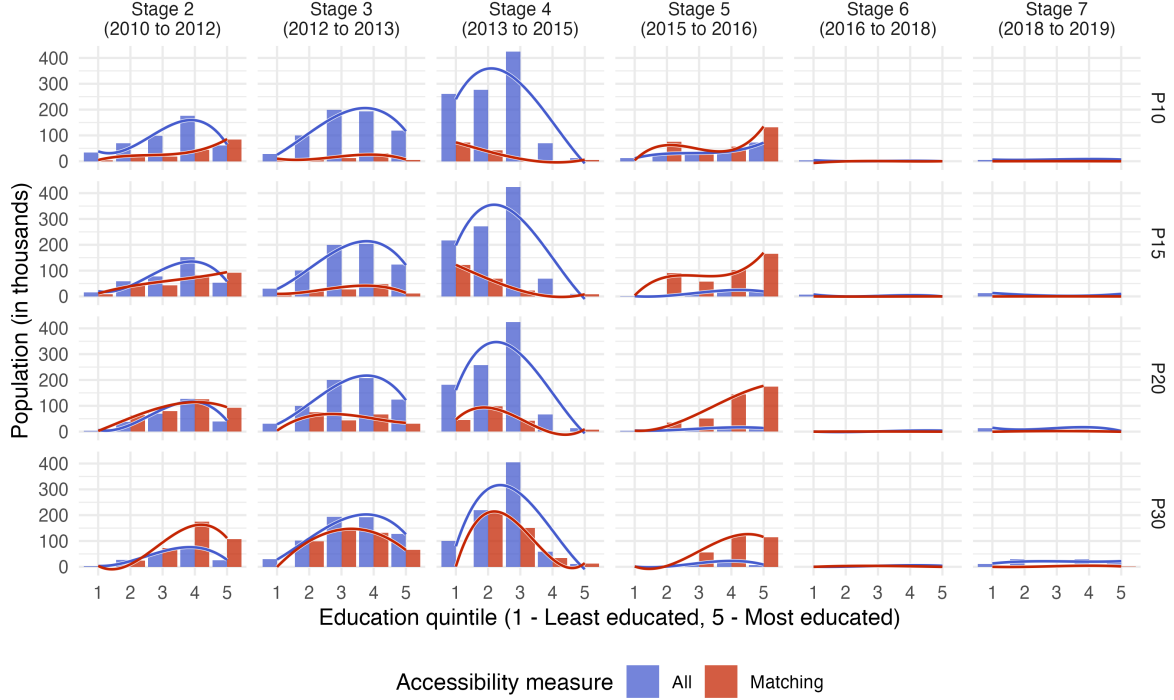


Figure 8: Equity analysis by education level over various temporal stages.

<!-- ... -->

to lower-educated groups, between the first and third quintile. This makes sense, since the MXB-LIII enabled direct connection to the MPTN to two densely populated municipalities characterised by high-deprivation levels, i.e. *Ciudad Nezahualcóyotl* and *Chimalhuacan*. However, these effects diminish at higher accessibility thresholds, likely because low-educated populations reside in less central areas. Thus, as the threshold increases, lower-educated zones fall short, while areas with higher education levels are considered beneficiaries.

Figure 8 also highlights the matching employment benefits in Stage 5 (2015-2016), as discussed earlier. Interestingly, these benefits primarily went to the more educated population between the third and fifth quintiles, a pattern consistent across all thresholds. As noted, no significant population was taken out of the minimal accessibility thresholds in Stages 6 and 7.

5 Discussion

The results examining the spatial differences between generic and matching accessibility measures showed important interactions with the MAUP. Specifically, the larger the scale, the less relevant the distinction of a matching accessibility is, as shown by the increased correlation. To an extent, this is explained by the tendency of larger units to emphasize within-zone opportunities, given the increased travel costs for residents to leave the unit. Thus, at large scales, accessibility measures tend to reflect employment density. Interestingly, the post code retained matching employment information to a very good extent, as shown by

the low correlation. By implication and as shown visually, the role of the public transport infrastructure is gradually lost with the use of large scales too. Similarly, post code units showed to retain some of these features. Analysts who aim to account for matching characteristics and the role of transport infrastructure should take these effects into consideration.

Furthermore, our results indicate that zones that showed significant similarity between the ‘All’ and ‘Matching’ measures were predominantly composed of the most educated population. This reflects a West-East divide, aligning with the well-documented spatial inequalities in GMC, e.g. related to household income and car ownership (**Guerra2015?**), healthy food environments (**Ortega-Avila2023?**), and social mix (**Letouze2022?**). From the urban economic perspective, high-educated populations are likely to be in a position to afford residential locations that maximise accessibility to employment opportunities that they consider valuable, outbidding less affluent groups (**Duranton2015?**). An implication for accessibility analyses which draw on the assumption of equivalent attractiveness of employment might systematically overestimate the relevant opportunities available to less educated populations. This is expected to be more relevant in context with higher economic inequalities.

The results regarding the evaluation of public transport infrastructure over time are varied. Firstly, it is confirmed that a simple measure alone, such as the global average of accessibility, might disregard key transport equity insights (**Karner2024?**). Specifically, while certain extensions of the MPTN reflect a substantial improvement on average, an equity analysis from a sufficiency perspective reflects negligible effects. We argue both indicators supplement each other. For instance, the results highlight a concentration of the benefits in Stage 7 among populations that were already served. This is consistent regardless of the accessibility measure being used.

Secondly, it is shown that at a higher level both accessibility measures point into similar directions with parallel trends. However, there are particularities to note. The matching accessibility measure was more sensitive to sufficiency thresholds in the equity analysis. This might be related to the observation that the distribution of the baseline to define the thresholds presents a more skewed distribution.

The comparison between measures also showed some complementarity. For example, the fact that the matching measure reflects that there are populations benefited by particular infrastructure interventions and the general does not, implies that certain populations might have received ‘general’ accessibility improvements previously and, later, specific ones about matching employment opportunities. Both indicators are useful when considering that the former measure as a form of *generalized access to urban opportunities*, as the location of employment often reflects the provision of a service (**Merlin2017?**; Levinson and Wu 2020). Meanwhile, a distinction of matching employment opportunities is useful when it is applied to commuting to work patterns or labour market inequalities, for example.

Thirdly, the particular differences arising from the inclusion of matching opportunities in equity analyses vary with a range of outcomes from case-to-case. It is shown that the implementation of a transport project can potentially imply benefits either for well educated or low-educated groups only, or measures can be consistent at times. In the area of study, this is observed to be related to the location of the transport improvement with respect of the overall transport network, and the demographic characteristics of the population in proximity to the public transport services. This reinforces the important role of accessibility measure in public transport evaluation, given that it is able to capture key elements of the transport system and characteristics of the population simultaneously. Thus, it is recommended to use both measures to form a comprehensive insights of transport infrastructure on a case-based basis.

5.1 Limitations and future work

The present study comes with the following implications. Specifically, the analyses assume that all population within-zones share similar demographic characteristics, e.g. same education levels or employment opportunities. This is a broader issue in geography discussed under the ‘Ecological Fallacy’ literature (**Walker2021?**). This is also related to limitations of data, given that individual-level data for both jobs and residents are restricted to maintain privacy. Also, the modelled matching access measure used assume limited social mobility. Namely, only equal education-income deciles are allowed. This can be partially validated by verifying general trends in the area of study. Still, additional research in different context can clarify the sensitivity of the assumption. Furthermore, the analyses focus on the MPTN only. This overlooks the street transit services. In the case of study, this is largely constrained by the information available, given that comprehensive structured timetable data is not available.

Future work is recommended to examine the sensitivity of results when using individual-level job characteristics to establish a match between residents and opportunities. Similar analysis in a diversity of context with a range of social and economic disparities would help to identify consistent patterns. This could help to establish the extent of the findings in our empirical case. Furthermore, further work should examine the assumptions of matching measures and their relevance for specific applications, e.g. in the housing market or commuting to work patterns.

6 Conclusions

The present study relaxed a common assumption of accessibility analysis that access to employment is equally attractive to all types of residents. The study examines the specific differences between a generic and matching accessibility measures taking into the account the MAUP, and its relevance for equity assessments of public transport infrastructure.

It is shown that both components of the modifiable areal unit problem, namely scale and zoning, play an important role in this distinction. Specifically, the added information regarding the matching employment is gradually smoothed with increased scale. Also, the difference between measures is sensitive to zoning as post codes, for example, can over-represent the role of large zones in local spatial analyses.

A sensitivity analysis is recommended particularly in context with high social inequality. This is because the allocation of residential land use might push low-income residents more acutely with the consequence of the level of access to relevant employment being systematically underestimated for less advantaged populations. Also, the interpretation and information provided by the distinct measures will depend on specific applications. In this study, we have shown that both measures play complementary roles.

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